UNIRATIONAL VARIETIES THAT ARE NOT RATIONAL

[after M. Artin and D. Mumford]

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This report contains a description of the variety constructed by Artin and Mumford, together with the proof that it is unirational but not rational. It also states the theorems proved by Clemens and Griffiths [4], and by Manin and Iskovskih [6], which yield other examples. I attended talks by Artin and Mumford on the same subject, and I have drawn heavily on them.

Throughout, the expression "algebraic variety" means "a separated scheme of finite type over $\mathbf{C}$, reduced and irreducible." We write $k(X)$ for the field of rational functions on an algebraic variety X.

1. Rationality and unirationality

Let K be a finitely generated extension of $\mathbf{C}$. Then there exists an algebraic variety X such that K is isomorphic to $k(X)$. One calls X a *model* of K. By [5], K always admits a smooth projective model.

Let K and L be finitely generated extensions of $\mathbf{C}$, with models X and Y. Homomorphisms $a^*\colon K \to L$ correspond bijectively to dominant rational maps $a\colon Y \dashrightarrow X$, by the rule $a^*(f) = f \circ a$. By [5], if X and Y are smooth and projective, every rational map $a\colon Y \dashrightarrow X$ admits a factorization

(1)
$$a = b \circ c_n^{-1} \circ \cdots \circ c_1^{-1}$$

where b and the c_i are everywhere defined morphisms and where Y_{i+1} is obtained from Y_i by blowing up a smooth subvariety Z_i of codimension $\geqslant 2$.

If $a^*\colon K \to L$ is an isomorphism, one can apply the preceding result to (a^{*-1}, Y_{n+1}, X). Iterating, one obtains a commutative diagram of scheme morphisms

(2)
$$\begin{tikzcd}
 & Y'_n \arrow[r, "y_n"] \arrow[d, "a_n"] & \cdots \arrow[r, "y_2"] & Y'_1 \arrow[r, "y_1"] \arrow[d, "a_1"] & Y \arrow[d, "a"] \\
X & X'_n \arrow[l, "x_n"'] & \cdots \arrow[l, "x_2"'] & X'_1 \arrow[l, "x_1"'] & X \arrow[l]
\end{tikzcd}$$

In this diagram, the horizontal arrows x_i, y_i are composites of blow-ups with smooth centers, as in (1); the a_i and b_i are birational.

An invariant of a smooth projective variety X is said to be *birational* if it depends only on $k(X)$. One often verifies birational invariance using (2) (for a typical example, see Proposition 2.1). Invariants of a field K are usually defined as the value of some birational invariant of any smooth projective model of K.

Definition 1.1. Let X be an algebraic variety of dimension n. One says that X is *unirational* if the following equivalent conditions hold:

(i) $k(X)$ is a subfield of a purely transcendental extension $\mathbf{C}(T_1, \ldots, T_r)$ of $\mathbf{C}$;
(ii) a finite extension of $k(X)$ is isomorphic to $\mathbf{C}(T_1, \ldots, T_n)$;
(iii) if X is smooth and projective, there exists a diagram (1) with $Y = \mathbf{P}^r(\mathbf{C})$;
(iv) if X is smooth and projective, there exists a diagram (1) with $Y = \mathbf{P}^n(\mathbf{C})$.

One may suppose X smooth and projective, and one has

$$\text{(ii)} \Rightarrow \text{(i)} \Rightarrow \text{(iii)} \Rightarrow \text{(iv)} \Rightarrow \text{(ii)}.$$

Definition 1.2. One says that X is *rational* if, equivalently,

(i) $k(X)$ is a purely transcendental extension of $\mathbf{C}$;
(ii) if X is smooth and projective, there exists a diagram (2) with $Y = \mathbf{P}^n(\mathbf{C})$.

The most obvious birational invariants do not distinguish unirational varieties from rational varieties.

Proposition 1.3. *Let X be a smooth projective unirational algebraic variety of dimension $n > 0$.*

(i) *The plurigenera $P_k = \dim H^0(X, (\Omega_X^n)^{\otimes k})$ $(k > 0)$ vanish. More generally, $H^0(X, (\Omega_X^1)^{\otimes k}) = 0$ for $k > 0$ (see* [9, Remark, p. 102]*).*
(ii) *X is simply connected* [10].

For $n = 1$ or 2, property (i) already characterizes rational varieties. Thus a unirational variety of dimension at most 2 is rational, and a subextension of $\mathbf{C}(T_1, \dots, T_k)$ of transcendence degree at most 2 is automatically purely transcendental. More precisely:

a) If $n = 1$ and $H^0(X, \Omega_X^1) = 0$, then $X \simeq \mathbf{P}^1(\mathbf{C})$. Indeed, by Riemann–Roch, for every $x \in X$, the line bundle $\mathcal{O}(x)$ defines an embedding of X into $\mathbf{P}^1(\mathbf{C})$.

b) For $n = 2$, the Castelnuovo–Enriques criterion says that X is rational if $P_a = P_2 = 0$, i.e. if $H^0(X, \Omega_X^2) = H^0(X, (\Omega_X^2)^{\otimes 2}) = 0$; see [9].

For $n > 2$, it has been very difficult to define or compute birational invariants capable of distinguishing rational from unirational varieties.

2. A few birational invariants

A) Artin and Mumford.

Proposition 2.1. *The torsion subgroup $H^3(X, \mathbf{Z})_{\mathrm{tors}}$ of $H^3(X, \mathbf{Z})$ is a birational invariant of the smooth projective variety X.*

If X_2 is obtained from a smooth variety X_1 by blowing up a smooth subvariety $Z \subset X_1$ of codimension $r \geqslant 2$, then (SGA 5, VII)

$$H^n(X_2, \mathbf{Z}) \simeq H^n(X_1, \mathbf{Z}) \oplus \bigoplus_{i=1}^{r-1} H^{n-2i}(Z, \mathbf{Z}).$$

In particular, $H^3(X_2, \mathbf{Z}) \simeq H^3(X_1, \mathbf{Z}) \oplus H^1(Z, \mathbf{Z})$. Since $H^1(Z, \mathbf{Z})$ is torsion-free, one gets

$$H^3(X_1, \mathbf{Z})_{\mathrm{tors}} \xrightarrow{\sim} H^3(X_2, \mathbf{Z})_{\mathrm{tors}}.$$

Assume $k(X) \simeq k(Y)$, and choose a diagram (2). One obtains

$$\begin{tikzcd}
H^3(X,\mathbf{Z})_{\mathrm{tors}} \arrow[r, "\sim"] & H^3(X'_1,\mathbf{Z})_{\mathrm{tors}} & H^3(X'_n,\mathbf{Z})_{\mathrm{tors}} \arrow[l] \arrow[r, "\sim"] & H^3(X,\mathbf{Z})_{\mathrm{tors}} \\
H^3(Y,\mathbf{Z})_{\mathrm{tors}} \arrow[u, "a_1^*"] \arrow[r, "\sim"] & H^3(Y'_1,\mathbf{Z})_{\mathrm{tors}} \arrow[u, "b_1^*"'] & H^3(Y'_n,\mathbf{Z})_{\mathrm{tors}} \arrow[u, "b_n^*"'] \arrow[l] \arrow[r, "\sim"] & H^3(Y,\mathbf{Z})_{\mathrm{tors}} \arrow[u, "a_2^*"']
\end{tikzcd}$$

Since $a_2^* \circ b_n^*$ is bijective, b_n^* is injective. Since $b_n^* \circ a_n^*$ is bijective and b_n^* is injective, a_n^* is bijective, and the assertion follows.

Construction 2.2. We shall construct a smooth projective unirational variety X of dimension 3 with $H^3(X, \mathbf{Z})_{\mathrm{tors}} \neq 0$ (in fact, $H^3(X, \mathbf{Z}) \simeq \mathbf{Z}/(2)$). Multiplying this variety by $\mathbf{P}^r(\mathbf{C})$ yields an analogous example in every dimension $\geqslant 3$.

B) Clemens and Griffiths. Let X be a smooth projective variety of dimension 3 such that $H^0(X, \Omega^3_X) = 0$. Then the Hodge decomposition of $H^3(X, \mathbf{C})$ reduces to

$$H^3(X, \mathbf{C}) = H^{2,1} \oplus H^{1,2},$$

and, by Weil [11], the complex torus

$$J(X) = H^3(X, \mathbf{Z}) \backslash H^3(X, \mathbf{C}) / H^{2,1}$$

is an abelian variety: the *intermediate Jacobian* of X. Let

$$\Phi : H^3(X, \mathbf{Z}) \otimes H^3(X, \mathbf{Z}) \to H^6(X, \mathbf{Z}) \simeq \mathbf{Z}$$

be the cup product. By Poincaré duality, the induced alternating form on $H^3(X, \mathbf{Z})/\text{torsion}$ has discriminant one. If $H^1(X, \mathbf{Z}) = 0$, it follows from Weil [11] that this defines a principal polarization on $J(X)$.

Proposition 2.3 (. *4] If X is rational, the principally polarized abelian variety $J(X)$ is a product of Jacobians.*

In the sketch of proof below, the sign $\simeq$ denotes an isomorphism of principally polarized abelian varieties.

a) $J(\mathbf{P}^3(\mathbf{C})) = 0$.

b) If Y'' is obtained from a smooth projective variety Y' by blowing up a smooth curve Z with Jacobian $J(Z)$ (resp. a point), one has [4, 3.11]

$$J(Y'') \simeq J(Y') \times J(Z)$$

(resp. $J(Y'') \simeq J(Y')$).

c) If a morphism $b\colon Y' \to X$ is birational, then $J(X)$ is a direct factor of $J(Y')$: there exists a principally polarized abelian variety B from the main series such that

$$J(Y') \simeq J(X) \times B.$$

d) Let $(A_i)_{1 \leqslant i \leqslant n}$ be nonzero principally polarized abelian varieties. Assume that none of the A_i admits a decomposition $A_i \simeq A'_i \times A''_i$ (for example, assume A_i is a Jacobian). Then any decomposition of $A = \prod A_i$ is of the form

$$A \simeq \left(\prod_{i \in I} A_i\right) \times \left(\prod_{i \notin I} A_i\right) \quad \text{for } I \subset [1, n]$$

([4, 3.23]).

If X is rational, there exists a diagram (2) with $Y = \mathbf{P}^3(\mathbf{C})$ and b birational. By a) and b), $J(Y'_{n+1})$ is a product of Jacobians. By c) and d), so is $J(X)$.

Now take for X a smooth cubic hypersurface in $\mathbf{P}^4(\mathbf{C})$. One has

$$H^1(X, \mathbf{Z}) = H^3(X, \mathbf{Z})_{\text{tors}} = H^0(X, \Omega^1_X) = 0,$$
$$\dim J(X) = \dim H^1(X, \Omega^2_X) = 5,$$

and X is unirational by [7]. Let S be the surface parametrizing the lines contained in X, and let $\alpha\colon S \to \text{Alb}(S)$ be the Albanese map. One of the essential results of [4] is the following.

Theorem 2.4 (. *4] There exists an isomorphism $J(X) \xrightarrow{\sim} \text{Alb}(S)$ carrying the divisor Θ of $J(X)$ to the image of $S \times S$ under the map $(x, y) \mapsto \alpha(x) - \alpha(y)$.*

Clemens and Griffiths then use results from [1] to prove that $(\text{Alb}(S), \theta(S \times S))$ cannot be the polarized Jacobian of a curve. Another proof of the irrationality of X is suggested in the appendix to [4]; it amounts to proving the following.

Theorem 2.5. *The singular locus of the divisor Θ of $J(X)$ has dimension 0, hence codimension > 4. In fact it is reduced to the image under θ of the diagonal of $S \times S$.*

This is known never to happen for the theta divisor of a Jacobian.

C) Manin and Iskovskih. The group of birational automorphisms of a variety X (equivalently, the group of $\mathbf{C}$-automorphisms of $k(X)$) is a birational invariant of X. It is this invariant that Manin and Iskovskih use to show that a smooth quartic hypersurface in $\mathbf{P}^4(\mathbf{C})$ is never rational.

Theorem 2.6 ([6])**.** *Let X and Y be smooth quartic hypersurfaces in $\mathbf{P}^4(\mathbf{C})$. Every birational isomorphism $a\colon X \dashrightarrow Y$ is automatically biregular.*

Since some quartic hypersurfaces are unirational [7], this theorem provides a new example of unirational but nonrational varieties.

3. The Artin–Mumford example

Consider in $\mathbf{P}^2(\mathbf{C})$ a configuration consisting of

a) two smooth cubic curves C_1 and C_2, with equations $f_1 = 0$ and $f_2 = 0$, meeting transversally;

b) a smooth conic Q, with equation $q = 0$, meeting each C_i at three distinct tangency points.

We shall construct

a) a rank-3 vector bundle V on $\mathbf{P}^2(\mathbf{C})$;
b) a quadratic form $\Phi\colon V \to \mathcal{O}$ ($\mathcal{O}$ an invertible sheaf) with discriminant $f_1 f_2$ (if Δ, a section of $\mathcal{L}^{\otimes 3} \otimes (\Lambda^3 V)^{\otimes(-2)}$, denotes the discriminant, then the subscheme of $\mathbf{P}^2(\mathbf{C})$ cut out by $\Delta = 0$ is $C_1 \cup C_2$).

Let $X \subset \mathbf{P}(V^*)$ be the conic bundle over $\mathbf{P}^2(\mathbf{C})$, degenerating along $C_1 \cup C_2$, defined by the homogeneous equation $\Phi = 0$. The following conditions will hold.

c) Φ has rank $\geqslant 2$ at every point: for $s \in C_1 \cup C_2$, the fiber $X_s = p^{-1}(s)$ is the union of two concurrent lines; the only singular points of X are the singular points of the fibers X_s for $s \in C_1 \cap C_2$.

d) Let S be the double cover of $\mathbf{P}^2(\mathbf{C})$ branched along Q, and write $X_S = X \times_{\mathbf{P}^2(\mathbf{C})} S$. Then X_S/S admits a section.

e) For $s \in (C_1 \cup C_2) - Q$, the two points $\varphi(t)$ with $\varphi(t) = s$ lie on the two irreducible components of X_s.

Locally on $C_1 \cup C_2$ (for the usual topology), the restriction of the bundle X to $C_1 \cup C_2$ is obtained by gluing two $\mathbf{P}^1$-bundles along a section. However, when one goes around a loop in $C_1 \cup C_2$, these two bundles may be interchanged: there is an unramified double covering $(C_1 \cup C_2)^*$ of $C_1 \cup C_2$ whose two local sheets correspond locally to the two $\mathbf{P}^1$-bundles whose union is $X|_{C_1 \cup C_2}$. Let $\pi_i\colon C_i^* \to C_i$ be the double covering of C_i induced by $(C_1 \cup C_2)^*$.

Lemma 3.1. *C_i^* is irreducible.*

By e), C_i^* is the normalization of the double cover of C_i induced by S. The lemma follows from the fact that the section q of $\mathcal{O}(2)|_{C_i}$ is not the square of a section of $\mathcal{O}(1)|_{C_i}$; indeed, if q were a square, the three intersection points of Q with C_i would be collinear.

The double covering $C_i^* \to C_i$ corresponds to an order-2 character of the fundamental group of C_i and to a local system A_i, locally isomorphic to $\mathbf{Z}$, defined by the exact sequence of sheaves on C_i

$$0 \longrightarrow A_i \longrightarrow \pi_{i*}\mathbf{Z} \xrightarrow{\mathrm{Tr}} \mathbf{Z} \longrightarrow 0, \tag{3}$$

where Tr is "the sum of the values on the two sheets of the covering."

One has

$$H^0(C_i, A_i) = 0, \qquad H^1(C_i, A_i) \simeq \mathbf{Z}/2, \qquad H^2(C_i, A_i) = \mathbf{Z}/2. \tag{4}$$

Lemma 3.2. *X is unirational.*

The surface S is a quadric in $\mathbf{P}^3(\mathbf{C})$, hence is rational. Since X_S/S admits a section, there is a birational isomorphism $X_S \dashrightarrow S \times \mathbf{P}^1(\mathbf{C})$, and X is an image of the rational variety X_S.

The nine singular points of X are ordinary quadratic points. One resolves each of them as follows (cf. [3]).

a) Blow it up. This resolves X and replaces the singular point by a quadric P, isomorphic to $\mathbf{P}^1(\mathbf{C}) \times \mathbf{P}^1(\mathbf{C})$.

b) Choose a projection $P \to \mathbf{P}^1(\mathbf{C})$, and contract inside the blow-up $\widetilde{X}$ of X the fibers of this projection. This is possible because $\widetilde{X}$ is smooth, its fibers are isomorphic to $\mathbf{P}^1(\mathbf{C})$, and the normal bundle to P in $\widetilde{X}$ induces on each fiber a line bundle $\mathcal{O}(-1)$ [8].

Let $\bar{X}$ be the resulting variety. There is a morphism from $\bar{X}$ to X, and $\bar{X}$ is obtained from X by replacing each singular point by a projective line.

The fibers of the projection $p\colon \bar{X} \to \mathbf{P}^2(\mathbf{C})$ are of the following three kinds:

$s \notin C_1 \cup C_2$: a conic;
$C_1 \cup C_2$, $s \notin C_1 \cap C_2$: two concurrent lines;
$s \in C_1 \cap C_2$: three lines arranged as indicated.

When one specializes from the generic point ξ of $\mathbf{P}^2(\mathbf{C})$ to the generic point η_i of C_i ($i = 1, 2$), and then to a point $s \in C_1 \cap C_2$, the irreducible components specialize as indicated in the following diagram:

$$(3.4) \qquad \begin{array}{ccccccccc} \eta_1 & & & & \xi & & & & \eta_2 \\ & \diagdown & & \diagup & & \diagdown & & \diagup & \\ & & \times & & & & \times & & \\ & \diagup & & \diagdown & & \diagup & & \diagdown & \\ \eta_2 & & & & \eta_1 & & & & \eta_2 \end{array} \tag{5}$$

(the crosses represent irreducible components; all multiplicities are one). The rather tedious verification of this local result will be omitted. One deduces that $R^0p_*\mathbf{Z} = \mathbf{Z}$, $R^1p_*\mathbf{Z} = 0$, and the following exact sequence of sheaves.

Lemma 3.3. *One has*

$$0 \longrightarrow A_1 \oplus A_2 \longrightarrow R^2p_*\mathbf{Z} \xrightarrow{\mathrm{Tr}} \mathbf{Z} \longrightarrow 0.$$

One can now use the Leray spectral sequence of p to compute $H^*(\bar{X}, \mathbf{Z})$. The essential fact is the following, which implies the irrationality of X.

Lemma 3.4. *One has $H^3(\bar{X}, \mathbf{Z}) = \mathbf{Z}/(2)$.*

Let us compute the E_2-terms of the Leray spectral sequence of p. Since $R^0p_*\mathbf{Z} = \mathbf{Z}$, we have $E_2^{p,0} = \mathbf{Z}, 0, \mathbf{Z}, 0, \mathbf{Z}$ for $p = 0$ to 4; and since $R^1p_*\mathbf{Z} = 0$, we have $E_2^{p,1} = 0$.

To compute $E_2^{p,2} = H^p(R^2 p_* \mathbf{Z})$, one uses the long exact sequence attached to the short exact sequence of sheaves in Lemma 3.3. One finds

$$0 \longrightarrow E_2^{0,2} \xrightarrow{\alpha} \mathbf{Z} \longrightarrow E_2^{1,2} \longrightarrow 0,$$

$$0 \longrightarrow \mathbf{Z}/2 \oplus \mathbf{Z}/2 \longrightarrow E_2^{2,2} \longrightarrow 0,$$

$$E_2^{3,2} = 0 \qquad \text{and} \qquad E_2^{4,2} = \mathbf{Z}.$$

One checks easily that $\alpha(E_2^{0,2}) = 2\mathbf{Z}$ (this is not in fact necessary for the proof that $H^3(\bar{X}, \mathbf{Z})_{\mathrm{tors}} \neq 0$). Thus the $E_2^{p,q}$ are as follows:

$$\begin{array}{c|ccccc}
q & & & & & \\
\hline
2 & \mathbf{Z} & \mathbf{Z}/2 & E_2^{2,2} \supset 0 & 0 & \mathbf{Z} \\
1 & 0 & 0 & 0 & 0 & 0 \\
0 & \mathbf{Z} & 0 & \mathbf{Z} & 0 & \mathbf{Z} \\
\hline
 & 0 & 1 & 2 & 3 & 4 \quad p
\end{array}$$

The differentials d_r can only vanish, so $E_2^{p,q} = E_\infty^{p,q}$, and the lemma follows.

One sees that it was essential to have two curves $C_i \subset \mathbf{P}^2(\mathbf{C})$ along which X degenerates, because one kills a $\mathbf{Z}/2$ in passing from

$$H^1\big(\mathrm{Ker}(R^2 p_* \mathbf{Z} \to \mathbf{Z})\big) = H^1(R^2 p_* \mathbf{Z}) = E_2^{1,2}$$

to $H^1(R^2 p_* \mathbf{Z}) = E_2^{1,2}$.

It remains to carry out the promised constructions. Artin and Mumford are able to prove *a priori* the existence of a conic bundle of the required type. I shall content myself with giving equations that define one.

1) The image of $f_1 f_2$ in $H^0(Q, \mathcal{O}(6))$ is the square of an element $g \in H^0(Q, \mathcal{O}(3))$, because Q is rational and the zeros of $f_1 f_2$ on Q are double. This g lifts to an element still denoted $g \in H^0(\mathbf{P}^2(\mathbf{C}), \mathcal{O}(3))$ because $H^1(\mathbf{P}^2(\mathbf{C}), \mathcal{O}(1)) = 0$. Since $f_1 f_2 - g^2$ vanishes on Q, it is a multiple of q:

$$f_1 f_2 = g^2 + qd.$$

2) Take $V = \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O}$, and the quadratic form

$$\Phi \colon V \to \mathcal{O} \colon (x, y, z) \longmapsto qx^2 + 2gxy - dy^2 - z^2.$$

Then $\Delta = g^2 + qd = f_1 f_2$. At a point where Φ had rank $\leqslant 1$, one would have $\Delta = q = d = 0$, hence $g = 0$, and $\Delta = g^2 + qd$ would have a double zero. This is impossible because $C_1 \cap C_2 \cap Q = \varnothing$. Finally, S is identified with the subscheme of X cut out by the equation $y = 0$, and d) and e) follow.

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